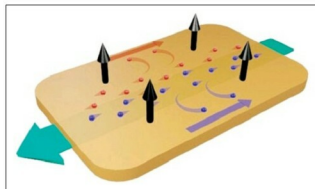


the first flexible batteries inside wearable electronics. The secret to these batteries is a traditional Japanese fold-and-cut practice of *kirigami* (which derives from origami). *Kirigami* is a compound word made of two words, *kiru* (to cut) and *kami* (paper). The Japanese practice involves making small slits or paper crease cuts and folding the paper remnants to create some form of design.

In this case, the ASU team, led by ASU mechanical and aerospace engineering associate professor Hanqing Jiang, created cut-and-twist patterns in the creases of Reynolds Wrap aluminium foil, coated the foil with electrode-laden, conductive material and then replaced the 300mAh battery inside Samsung's Gear 2 smartwatch with the coated kirigami-cut aluminium foil.

Material with magnetoresistance for better electronics

Researchers in Germany have found a material that could be used to build smaller and faster electronics. Known as niobium-phosphide, the material's elec-



Niobium-phosphide for building smaller and faster electronics
(Image courtesy: www.sciencealert.com)

tronic resistance increases dramatically when exposed to a magnetic field.

The property called giant magnetoresistance is crucial to achieving large storage capacities that we have come to enjoy in our hard drives.

The material has an incredible magnetoresistance because its electrons are superfast, with a top speed of around 300km/s. In a magnetic field, this makes them vulnerable to a phenomenon known as Lorentz force, which causes an increasing percentage of electrons to flow in the wrong direction as the magnetic field becomes stronger.

Because of this force, extremely large magnetoresistance of 250 per cent was observed at room temperature. In comparison, materials that are currently used in hard drives usually exhibit magnetoresistance of between 40 per cent and 110 per cent at room temperature.

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